Flutter

Google Play's 16 KB Page Size Issue Resolved !!

⚠️Before

app-release.apk

com.abym.garments_hub (Version Name: 1.0.5, Version Code: 5)

APK size: 82.7 MB, Download Size: 40.4 MB

File	Size	Download Size	% of Download Size	Compressed	Alignment
	82 MB	40.3 MB	100%		APK does not support 16 KB devices
lib	73.2 MB	32.1 MB	79.5%		
x86_64	25 MB	10.4 MB	25.8%		
libflutter.so	11.2 MB	4.8 MB	12%	No	4 KB zip alignment, but 16 KB is required
libapp.so	8.1 MB	3.2 MB	8%	No	16 KB zip|64 KB LOAD section
libbarhopper_v3.so	5.6 MB	2.3 MB	5.7%	No	4 KB zip alignment, but 16 KB is required
libimage_processing_util_jni.so	45.1 KB	20.4 KB	0%	No	4 KB LOAD section alignment, but 16 KB is required
libdatastore_shared_counter.so	6.1 KB	2.3 KB	0%	No	16 KB
arm64-v8a	23 MB	10 MB	24.7%		
libflutter.so	10.2 MB	4.8 MB	11.8%	No	16 KB zip|64 KB LOAD section
libapp.so	8.1 MB	3.2 MB	7.9%	No	4 KB zip alignment, but 16 KB is required
libbarhopper_v3.so	4.7 MB	2 MB	5%	No	4 KB zip alignment, but 16 KB is required
libimage_processing_util_jni.so	28.3 KB	14.3 KB	0%	No	4 KB LOAD section alignment, but 16 KB is required
libdatastore_shared_counter.so	6.9 KB	2.6 KB	0%	No	4 KB zip alignment, but 16 KB is required
armeabi-v7a	19.2 MB	9.3 MB	23.1%		
libflutter.so	7.2 MB	4.1 MB	10.1%	No	16 KB
libapp.so	8.9 MB	3.5 MB	8.7%	No	16 KB
libbarhopper_v3.so	3.1 MB	1.7 MB	4.2%	No	4 KB
libimage_processing_util_jni.so	21.2 KB	12.6 KB	0%	No	16 KB
libdatastore_shared_counter.so	4.3 KB	2 KB	0%	No	4 KB
x86	5.9 MB	2.4 MB	5.9%		
libbarhopper_v3.so	5.8 MB	2.4 MB	5.9%	No	4 KB
libimage_processing_util_jni.so	34.5 KB	16.4 KB	0%	No	4 KB
libdatastore_shared_counter.so	5 KB	2.3 KB	0%	No	4 KB

✅After

gradle-wrapper.properties | settings.gradle | build.gradle | pubspec.yaml | app-release.apk | kdrop_down.dart | api_routes.da

com.abym.garments_hub (Version Name: 1.0.7, Version Code: 7)

APK size: 77.2 MB, Download Size: 37.9 MB

File	Size	Download Size	% of Download Size	Compressed	Alignment
	76.5 MB	37.9 MB	100%		
lib	68.3 MB	30.3 MB	79.9%		
x86_64	25.4 MB	10.6 MB	28%		
libflutter.so	11.6 MB	5 MB	13.3%	No	16 KB zip|64 KB LOAD section
libapp.so	8.1 MB	3.2 MB	8.6%	No	16 KB zip|64 KB LOAD section
libbarhopper_v3.so	5.6 MB	2.3 MB	6.1%	No	16 KB
libimage_processing_util_jni.so	47 KB	22.2 KB	0.1%	No	16 KB
libdatastore_shared_counter.so	6.1 KB	2.3 KB	0%	No	16 KB
libsurface_util_jni.so	4.8 KB	1.7 KB	0%	No	16 KB
arm64-v8a	23.3 MB	10.2 MB	26.9%		
libflutter.so	10.5 MB	4.9 MB	13%	No	16 KB zip|64 KB LOAD section
libapp.so	8.1 MB	3.2 MB	8.5%	No	16 KB zip|64 KB LOAD section
libbarhopper_v3.so	4.7 MB	2 MB	5.3%	No	16 KB
libimage_processing_util_jni.so	28.3 KB	14.4 KB	0%	No	16 KB
libdatastore_shared_counter.so	6.9 KB	2.6 KB	0%	No	16 KB
libsurface_util_jni.so	4.7 KB	1.7 KB	0%	No	16 KB
armeabi-v7a	19.6 MB	9.5 MB	25.1%		
libflutter.so	7.6 MB	4.2 MB	11.2%	No	16 KB
libapp.so	8.9 MB	3.5 MB	9.3%	No	16 KB
libbarhopper_v3.so	3.1 MB	1.7 MB	4.5%	No	16 KB
libimage_processing_util_jni.so	19.9 KB	12.3 KB	0%	No	16 KB
libdatastore_shared_counter.so	4.3 KB	2 KB	0%	No	16 KB
libsurface_util_jni.so	3.4 KB	1.5 KB	0%	No	16 KB
assets	6.2 MB	6 MB	15.8%		

⚡Upgrade Flutter SDK

Upgrade to the latest Flutter SDK v3.35.2

```
[✓] Flutter (Channel stable, 3.35.2, on Microsoft Windows [Version 10.0.26100.4946], locale en-IN)
[✓] Windows Version (Windows 11 or higher, 24H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
```

🚀Upgrade Flutter Packages

👉 Run this command to update your project's dependencies to the latest major versions

```
flutter pub upgrade --major-versions
```

⚡Upgrade Your Gradle Version

Upgrade the Gradle version to latest

Switch from Gradle 8.3 → 8.7 or latest

```
distributionUrl=https\://services.gradle.org/distributions/gradle-8.3-all.zip
```

↓

```
distributionUrl=https\://services.gradle.org/distributions/gradle-8.7-all.zip
```

⚡Upgrade Your AGP Version

Upgrade the AGP version 8.5.1 or higher

Check the official documentation at developer.android.com

```
plugins {
    id "dev.flutter.flutter-plugin-loader" version "1.0.0"
    //Update AGP Version From Here
    id "com.android.application" version "8.1.0" apply false
    id "org.jetbrains.kotlin.android" version "2.1.0" apply false
}
```

```
plugins {
    id "dev.flutter.flutter-plugin-loader" version "1.0.0"
    //Update AGP Version From Here
    id "com.android.application" version "8.6.0" apply false
    id "org.jetbrains.kotlin.android" version "2.1.0" apply false
}
```

⚡Update Your NDK Version

Update the NDK r28 or higher

NDK version r28 and higher compile 16 KB-aligned by default.

Check the official documentation at developer.android.com"

Update NDK version inside andriod/app/build.gradle

```
ndkVersion = flutter.ndkVersion
```

↓

```
ndkVersion = "28.0.12433566"
```

⚡Update Default Configuration

Update your default configuration according to your ndk version

To support compiling 16 KB-aligned shared libraries with Android NDK version r27 and higher, you need to update your ndk-build, build.gradle, build.gradle.kts, or linker flags as follows:

Check the official documentation at developer.android.com

Groovy

```
android {
    ...
    defaultConfig {
        ...
        // This block is different from the one you use to link Gradle
        // to your CMake or ndk-build script.
        externalNativeBuild {
            // For ndk-build, instead use the ndkBuild block.
            cmake {
                // Passes optional arguments to CMake.
                arguments "-DANDROID_SUPPORT_FLEXIBLE_PAGE_SIZES=ON"
            }
        }
    }
}
```

Kotlin

```
android {
    ...
    defaultConfig {
        ...
        // This block is different from the one you use to link Gradle
        // to your CMake or ndk-build script.
        externalNativeBuild {
            // For ndk-build, instead use the ndkBuild block.
            cmake {
                // Passes optional arguments to CMake.
                arguments "-DANDROID_SUPPORT_FLEXIBLE_PAGE_SIZES=ON"
            }
        }
    }
}
```

✅Check APK Support for 16KB Page

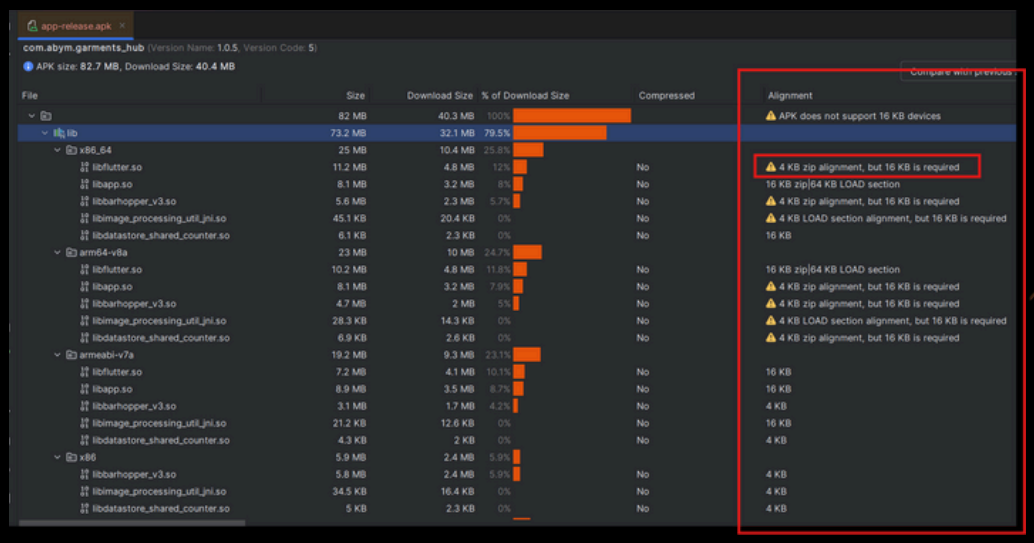
Build Flutter Release APK

From the menu bar, click Build > Analyze APK...

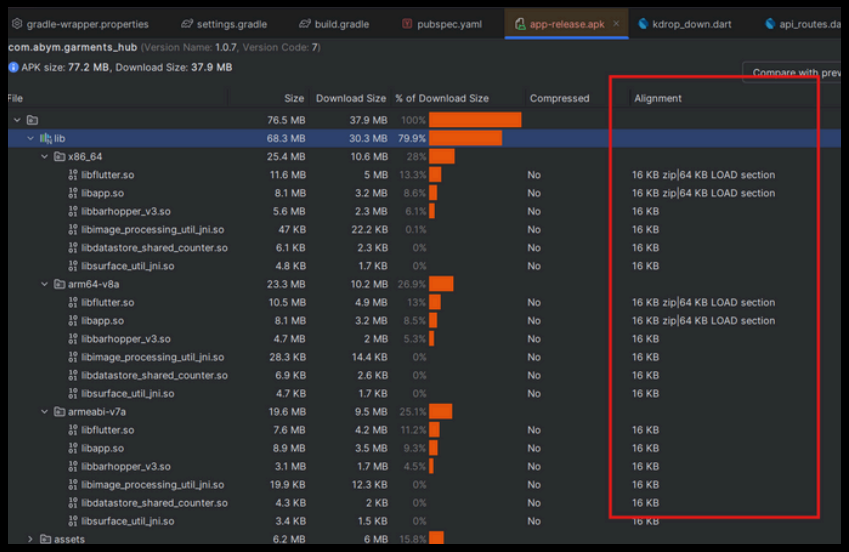
Choose the APK you want to analyze.

Look within the lib folder, which hosts shared **object (.so)** files if any are present. If any shared object files are present, your app uses native code. The Alignment column displays warning messages for any files that have alignment issues. If no shared object files are present or there is no lib folder, then your app doesn't use native code.

Before



After



✅Check APK Support for 16KB Page

- Go to Play Console → Test & Release → Latest releases and bundles
- Select Updated App Build → Check Memory page size

Before

App bundle: 5 (1.0.5)

Details

Releases	1 release Show detail
Supported Android devices	19,533 Go to device catalog
Localizations	86 localizations Show detail
Permissions	7 permissions Show detail
Features	4 features Show detail
Screen layouts	small, normal, large, xlarge
Native platforms	arm64-v8a, armeabi-v7a, x86, x86_64
API levels	23+
Target SDK	35
OpenGL ES versions	0.0+
OpenGL textures	No textures required
Memory page size	Does not support 16 KB Hide detail Learn more App does not support 16 KB
	Libraries that do not support 16 KB: base/lib/arm64-v8a/libimage_processing_util_jni.so base/lib/x86_64/libimage_processing_util_jni.so

After

6.aab (1.0.6)

Supported Android devices	18,154 Go to device catalog
Automatic protection	installer check
Localizations	86 localizations Show detail
Permissions	7 permissions Show detail
Features	4 features Show detail
Screen layouts	small, normal, large, xlarge
Native platforms	arm64-v8a, armeabi-v7a, x86_64
API levels	24+
Target SDK	36
OpenGL ES versions	0.0+
OpenGL textures	No textures required
Memory page size	Supports 16 KB You may still have errors if your app uses libraries that we couldn't detect, or if your code assumes device is using a specific page size. Learn more